

Overfitting and regularization

Module 5 - Lesson 3

Overview

Overfitting means memorizing training data instead of learning patterns, so test performance suffers. Regularization adds constraints that keep models simple.

Key Concepts

- Overfitting shows as low training loss but high validation loss.
- Underfitting is the opposite: high error everywhere; the model is too simple.
- More data, fewer parameters, and early stopping all fight overfitting.
- L1/L2 regularization penalize large weights, pulling them toward zero.
- Dropout randomly turns off neurons during training as a form of ensemble averaging.
- Data augmentation creates more training examples by transforming existing ones.

Example

```
import torch.nn as nn

model = nn.Sequential(
    nn.Linear(20, 64),
    nn.ReLU(),
    nn.Dropout(0.3),      # regularization
    nn.Linear(64, 1),
)

# Early stopping: stop when validation loss stops improving
```

Summary

Overfitting is memorization; regularization is the cure. Validation loss, dropout, weight penalties, early stopping, and more data keep models learning patterns instead of noise.

Practice Checklist

- Train a model and identify where train and validation loss diverge.
- Add dropout and compare validation performance.
- Use early stopping and report the epoch you stopped at.